PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
DAR ES SALAAM REGION
FORM SIX MOCK EXAMINATION
CHEMISTRY 3B

Code: 132/3B

Time: 3:00 Hours   Tuesday 20th February, 2024 A.M.

INSTRUCTIONS

1. This paper consists of **three (03)** questions. Answer **all** questions.
2. Question one carries **20 marks** and the other two carries **15 marks** each.
3. All answers must be written in the answer booklets provided.
4. Mathematical tables, and no – programmable calculators may be used.
5. Cellular phones and any other unauthorized materials are **not** allowed in the examination room.
6. Write your examination number on every page of your answer booklet(s)

   Atomic masses

   H = 1, O = 16, Na = 23, Cl = 35.5, C = 12, Mn = 55, K = 39, Cu = 63.5
   - Density of water = $1 g/cm^3$
   - Specifics heat capacity water = $4.18 J/g^oC$

This paper consists of 3 printed pages

1. You are provided with the following solutions

AA: A solution containing 1g of an impure sample of MnO_2 dissolve in 250cm³ of distilled water and 2.2g of oxalic acid crystals ($C_2H_2O_4 . 2H_2O$)

BB: A standard solution of 0.02M $KMnO_4$

CC: 2M H_2SO4 (Sulphuric acid)

Theory

Manganese dioxide and oxalic acid react quantitatively as follows

$MnO_2 + 2C_2H_2O_4 \rightarrow Mn^4 + 2H_2O + 4CO_2 \uparrow$

The excess oxalic acid reacts quantitatively with $KMnO_4$. As follows

$2MnO^-{}_4 + 5C_2H_2O_4 + 16H^+ \rightarrow 2Mn^{2+} + 8H_2O + 10CO_2$

Procedures

(i) Pipette 20cm³ (or 25cm³) of AA into a conical flask
(ii) Add about 20cm³ of dilute H_2SO_4 and heat the content of flask to about 80°C
(iii) Titrate the hot AA with BB in the burette until the purple colour no longer disappear
(iv) Repeat this produce two times and record your results in a tabular form as shown below.

Results

The volume of the pipette used___________cm³

Burette reading

Titration number	Pilot	1	2
Final reading (cm³)			
Initial reading (cm³)			
Title volume (cm³)			

Summary

For complete reaction ______________cm³ of AA required_____cm³ BB

Calculate

(a) Molarity of oxalic acid
(b) Concentration of both hydrated and unhydrated oxalic acid in g/dm³
(c) The percentage purity of the manganese dioxide in the sample.

2. You are provided with the following

A_1 : 5g of hydrated ammonium oxalate $(COONH_4)_2 . H_2O$
A_2 : 5g of anhydrous sodium carbonate
A_3 : Distilled water
: Stirrer / glass rod, thermometer.

Procedure:

(i) Record the room temperature
(ii) Pour 50cm³ of A_3 into a 100ml clean beaker
(iii) Record the temperature Tw.
(iv) Add 5g of A_2 into a beaker containing A_3. Stirr gently until the dissolution is complete record the temperature Ts.
(v) Clean a beaker and the thermometer then repeat the experiment using 5g of A1 and record Tw and Ts as in (iii) and (iv) above
(vi) Tabulate your result as shown below

Substance	V of water	Mass of salt	Tw: °C	Ts:°C	ΔT
Na_2Co_3					
$(COONH_4)_2 . H_2O$					

Questions.

(a) Calculate the molar mass of each salt
(b) Calculate the enthalpy change of solution of each salt
(c) Calculate the moral enthalpy change of solution of each salt
(d) State whether the process of dissolving each salt is endothermic or exothermic.
Given that the specific heat capacity of water = 4.18J/g°C.

3. Sample "A" is a sample salt containing one cation and one anion. Carefully carry out qualitative analysis experiment to identify ions present in the salt based on the following test.

(i) Appearance of the sample
(ii) Action of heat on sample
(iii) Flame test on a sample
(iv) Action of dil. HCl on a solid sample
(v) Solubility (divided the resulting solution into two portions)
(vi) To the first portion add few drops of NaoH solution. Then add in excess.
(vii) To the second portion add K1 solution warm and cool the mixture.
(viii) Transfer a small amount of the original sample into a test tube. Add copper turnings followed by concentrated sulphuric acid then warm.

Questions.

(a) Prepare a reverent table showing the qualitative analysis result.
(b) Write cation and anion present
(c) Write the formula of salt A.
(d) Write a balanced chemical equation for the reaction in experiment vi, vii

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MARKING SCHEME

(b) Concentration of hydrated oxalic acid.

From;

$$\text{Molarity} = \frac{\text{mass Concentration}}{\text{Molar mass}} \quad (00\tfrac{1}{2}\ \text{mark})$$

Mass Conc. = Molarity × Molar mass

But, Molar mass of $H_2C_2O_4 \cdot 2H_2O = 126\,g/mol$

Mass Conc. $= 0.02875\,moldm^{-3} \times 126\,gmol^{-1}$ ($00\tfrac{1}{2}$ mark)

$= 3.6225\,gdm^{-3}$ (01 mark)

Concentration of anhydrous oxalic acid.

$= 0.02875\,moldm^{-3} \times 90\,gmol^{-1}$

$= 2.5875\,gdm^{-3}$ (01 mark)

(c) Percentage purity of MnO_2 In the sample.

From;

$$\%\ \text{purity} = \frac{\text{Conc. of pure}}{\text{Conc. of Impure}} \times 100 \quad (01\ \text{mark})$$

$$\text{Concentration of Impure } MnO_2 = \frac{\text{mass}}{\text{Volume}}$$

$$= \frac{1g}{0.25dm^3}$$

$= 4g/dm^3$

(01 mark)

MnO_2 was finished at the end of reaction.

Thus; $a - x = 0$ and $x = 0.021$

Then; $a - 0.021 = 0$

$a = 0.021M$ (01 mark)

Molar mass of MnO_2 = 87 g/mol

Concentration (g/dm³) = Molarity × Molar mass

$= 0.021\,mol\,dm^{-3} \times 87\,g\,mol^{-1}$

$= 1.83\,g/dm^3$

Concentration of pure MnO_2 = $1.83\,g/dm^3$ (01 mark)

Then;

$$\%\ purity = \frac{1.83\,g/dm^3}{4\,g/dm^3} \times 100$$

$= 45.8\%$

∴ Percentage purity of MnO_2 in the Sample = 45.8% (01 mark)

Enthalpy change for $(COONH_4)_2 \cdot H_2O$

From; $\Delta H = -mc\Delta T$

But; $\rho = \frac{mass}{Volume}$

Given; $\rho_{H_2O} = 1g/cm^3$ and $V_{H_2O} = 50cm^3$

So, $Mass = \rho_{H_2O} \times V_{H_2O}$

$= 1g/cm^3 \times 50cm^3$

$= 50g$

Then; $\Delta H = -50g \times 4.18J/g°C \times -2°C$

$= +418J$

$\therefore$ Enthalpy change for $(COONH_4)_2 \cdot H_2O = +418J$ (01 mark)

(c) Molar enthalpy change for each salt;

For Na_2CO_3;

$q = \frac{\Delta H}{n}$ (01 mark)

But; $n = \frac{mass}{Molar\ mass} = \frac{5g}{106g/mol} = 0.047mol$ (01 mark)

$q = \frac{-836J}{0.047mol}$

$= -17787.234\ Jmol^{-1}$ (02 mark)

$\therefore$ Molar enthalpy change for $Na_2CO_3 = -17787.234Jmol^{-1}$

3.

S/N	Experiment	Observation	Inference
(i)	Appearance of solid sample.	Salt A was a white crystalline solid (01 mark)	- Non-trancition metals may be present ($00\frac{1}{2}$ mark) - NO_3^-, SO_4^{2-}, Cl^-, $C_2O_4^{2-}$, CrO_4^{2-}, NO_2^-, CH_3COO^-, $Cr_2O_7^{2-}$ may be present ($00\frac{1}{2}$ mark)
(ii)	Action of heat	Brown fumes evolved which turned moist blue litmus paper red. ($00\frac{1}{2}$ mark)	NO_3^- may be present except those of K^+, Na^+ and NH_4^+ ($00\frac{1}{2}$ mark)
(iii)	Flame test	Salt A burns with a blue flame ($00\frac{1}{2}$ mark)	Pb^{2+}, Sb^{2+} may be present ($00\frac{1}{2}$ mark)
(iv)	Action of dil. HCl	No gas evolved. ($00\frac{1}{2}$ mark)	SO_4^{2-}, Cl^-, NO_3^- May be present ($00\frac{1}{2}$ mark)
(V)	Solubility	Salt A was soluble in cold water ($00\frac{1}{2}$ mark)	Soluble salt was present ($00\frac{1}{2}$ mark)
(vi)	Addition of NaOH in small amount then in excess.	In small amount of NaOH, white precipitates formed. In excess NaOH the precipitates dissolved. ($00\frac{1}{2}$ mark)	Sn^{2+}, Pb^{2+} May be present. ($00\frac{1}{2}$ mark)
(vii)	Addition of KI solution then heated.	Yellow ppt. formed which disappeared on warming but reappear on cooling. ($00\frac{1}{2}$ mark)	Pb^{2+} Confirmed ($00\frac{1}{2}$ mark)
(viii)	Action of Copper turning to a solid sample followed by conc. H_2SO_4 then heated.	Brown fumes was evolved ($00\frac{1}{2}$ mark)	NO_3^- Confirmed. ($00\frac{1}{2}$ mark)